Title: From Words to Networks: How do LLM-Based Semantic Structures Predict Bilingual Codeswitching?

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Introduction: Codeswitching (CS)—the alternation between languages in bilingual speech—has been linked to the semantic structure of the bilingual lexicon. Xu et al. (2021) examined this relationship by calculating the clustering coefficient, a measure of how densely interconnected a word's semantic neighbors are. Their results show that codeswitches tend to occupy sparser semantic neighborhoods while their translation equivalents (TEs) lie in denser ones for switches from Spanish into English. However, this work relied on static word embedding models such as Word2Vec, which assign a single, context-independent vector to each word type. As a result, they may not capture variation in meaning across linguistic contexts, a limitation that is especially consequential for bilingual corpora where words are embedded in codeswitched discourse. LLMs overcome these limitations by generating contextualized token-level representations whose geometry adapts to the surrounding input, better capturing the moment-to-moment reconfiguration of bilingual semantic networks. Using network science approaches, we examine the organization of the bilingual lexicon, modeling both global and local structure: globally, we assess how the two languages interact within Word2Vec and LLM networks, and locally, we examine how word-level clustering shapes switching behavior.

Method: We analyzed the Tangrams Codeswitching Corpus, a dataset with 30,857 words produced by 90 Spanish-English bilinguals from Puerto Rico. For each unique word, we constructed semantic networks using (a) Word2Vec embeddings trained on the corpus and (b) XLM-RoBERTa embeddings fine-tuned on the same data. Pairwise cosine similarities were calculated from each model and transformed into weighted networks to capture semantic relationship strength. Globally, Louvain community detection (Blondel et al., 2008) was then used to partition networks into cohesive communities. Locally, each word's weighted clustering coefficient was computed from networks (Onnela et al., 2005). To test how clustering coefficient predicts CS, we conducted two-way ANOVAs with CS Direction (Spa→Eng vs. Eng→Spa) and Word Type (CS word vs. TE) as fixed factors.

Results: Community detection analyses revealed systematic differences in global network organization across models. The Word2Vec network showed strong cross-lingual segregation (Fig 1a), whereas the LLM network revealed substantial integration across languages (Fig 1b). Despite these global differences, both models showed the same local pattern: in Spanish to English switches, codeswitched words had lower clustering coefficients than their TEs (Fig 2).

Although the effect was much stronger in the Word2Vec network ($F(1, 944) = 784.51$, $p<0.001$), the LLM network also showed a similar trend ($F(1, 226) = 4.15$, $p=0.04$).

Conclusion: The models produced markedly different global network structures. The Word2Vec network consisted of two densely connected but largely segregated language-specific communities, whereas the transformer-based network formed a single integrated structure spanning both languages. This contrast may reflect differences in training data, as Word2Vec was trained only on our dataset, while the transformer was pretrained on large-scale multilingual data. Another possibility is that the models differ in how they represent meaning: Word2Vec assigns each word a single context-independent vector, whereas transformer embeddings are contextualized, which brings TEs into a common semantic space. Despite global differences, both models converged on the same local pattern with clustering increasing for switches into the Spanish but decreasing for switches into the English relative to their TEs. This asymmetry in clustering coefficient may reflect key differences in the types of switches. Spa→Eng switches are often lone insertions, which tend to involve lexical items with fewer tightly interconnected semantic neighbors, yielding lower scores. In contrast, switches into Spanish are more likely to occur as multiword stretches that require stronger semantic and syntactic integration (Pattichis et al., 2026), which favors items that are more densely connected in the semantic network and therefore show higher clustering coefficient.

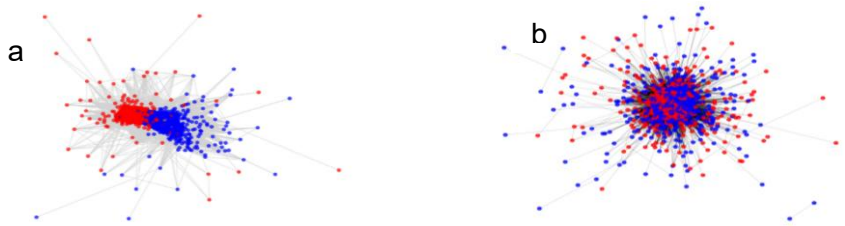

Figure 1 Community structure in bilingual lexical networks derived from different embedding models. Each node represents a word (blue = English; red = Spanish), and edges indicate semantic similarity between words. (a) Network constructed from Word2Vec embeddings shows a strongly segregated structure. (b) Network constructed from the fine-tuned XLM-R model shows a highly integrated structure.

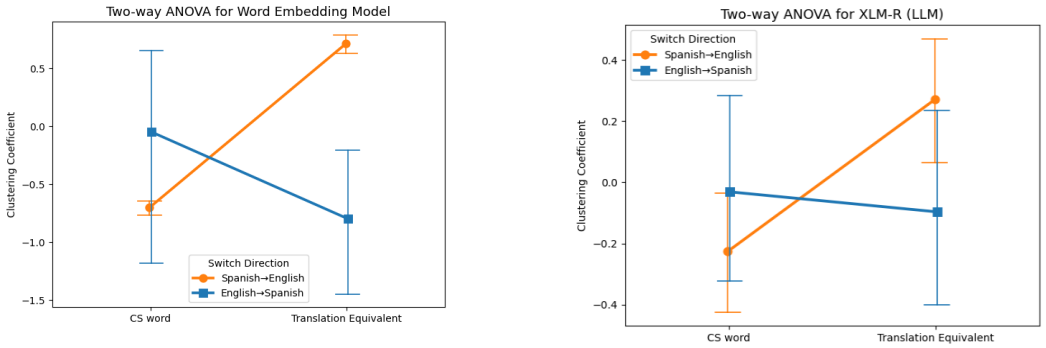

Figure 2 Clustering coefficients for code-switched words and their translation equivalents in networks built from Word2Vec (left) and LLM embeddings (right). Both models show a significant interaction: for Spanish→English switches (yellow), code-switched words have lower clustering coefficients than their translation equivalents, whereas for English→Spanish (blue) switches this pattern reverses. The effect is much stronger in the Word2Vec network, but the LLM network shows the same trend. Error bars indicate standard errors.

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